

Case Study Report

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
print("Displaying a part of dataset")
df = pd.read_csv('Indian_Climate_Dataset_2024_2025.csv')
df.head(10)
```

Displaying a part of dataset

	Date	City	State	Temperature_Max (°C)	\
0	2024-01-01	Mumbai	Maharashtra	32.5	
1	2024-01-01	Delhi	Delhi	25.4	
2	2024-01-01	Bengaluru	Karnataka	37.2	
3	2024-01-01	Chennai	Tamil Nadu	37.2	
4	2024-01-01	Kolkata	West Bengal	27.4	
5	2024-01-01	Hyderabad	Telangana	44.4	
6	2024-01-01	Ahmedabad	Gujarat	32.8	
7	2024-01-01	Jaipur	Rajasthan	40.4	
8	2024-01-01	Lucknow	Uttar Pradesh	42.3	
9	2024-01-01	Bhopal	Madhya Pradesh	27.4	

	Temperature_Min (°C)	Temperature_Avg (°C)	Humidity (%)	Rainfall (mm)	\
0	18.0	25.2	77.6	0.0	
1	10.7	18.1	84.1	0.0	
2	30.8	34.0	49.0	3.7	
3	30.4	33.8	34.2	9.5	
4	17.5	22.5	32.2	9.1	
5	31.6	38.0	91.1	51.9	
6	25.1	28.9	83.9	0.0	
7	33.5	37.0	30.4	8.2	
8	31.0	36.6	51.5	0.0	
9	15.3	21.3	79.5	0.0	

Wind_Speed (km/h)	AQI	AQI_Category	Pressure (hPa)	Cloud_Cover
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(%)				
0	3.3	259	Poor	1020.3
62.1				
1	9.0	130	Moderate	1008.4
46.0				
2	6.6	54	Satisfactory	1008.0
61.3				
3	9.0	176	Moderate	993.4
70.0				
4	9.2	97	Satisfactory	1008.2
56.9				
5	4.0	140	Moderate	996.9
9.3				
6	5.2	345	Very Poor	1018.1
12.1				
7	19.7	75	Satisfactory	992.6
39.1				
8	18.8	186	Moderate	1012.3
89.3				
9	14.0	73	Satisfactory	1005.0
7.4				

```

hottest_city = df.groupby('City')['Temperature_Max
(°C)'].mean().idxmax()
cleanest_city = df.groupby('City')['AQI'].mean().idxmin()
cloudy_city = df.groupby('City')['Cloud_Cover (%)'].mean().idxmax()

print(f"\nHottest City (Avg Max Temp): {hottest_city}")
print(f"City with Cleanest Air (Lowest Avg AQI): {cleanest_city}")
print(f"Cloudy city: {cloudy_city}")

```

```

Hottest City (Avg Max Temp): Kolkata
City with Cleanest Air (Lowest Avg AQI): Mumbai
Cloudy city: Lucknow

```

Thermal Peak: Kolkata recorded the highest average maximum temperatures.

Air Quality Champion: Mumbai achieved the lowest average Air Quality Index (AQI), categorizing it as the city with the cleanest air in this specific dataset.

Maximum Cloud Density: Lucknow experienced the highest average cloud cover.

```

print("Details of dataset:-\n")
print("Dataset Shape:", df.shape)
print("\nColumn Names and Types:")
print(df.info())

```

```

Details of dataset:-

```

Dataset Shape: (7310, 13)

Column Names and Types:

<class 'pandas.DataFrame'>

RangeIndex: 7310 entries, 0 to 7309

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	Date	7310 non-null	str
1	City	7310 non-null	str
2	State	7310 non-null	str
3	Temperature_Max (°C)	7310 non-null	float64
4	Temperature_Min (°C)	7310 non-null	float64
5	Temperature_Avg (°C)	7310 non-null	float64
6	Humidity (%)	7310 non-null	float64
7	Rainfall (mm)	7310 non-null	float64
8	Wind_Speed (km/h)	7310 non-null	float64
9	AQI	7310 non-null	int64
10	AQI_Category	7310 non-null	str
11	Pressure (hPa)	7310 non-null	float64
12	Cloud_Cover (%)	7310 non-null	float64

dtypes: float64(8), int64(1), str(4)

memory usage: 742.6 KB

None

Checking for missing values

print("\nMissing Values:\n")

print(df.isnull().sum())

Missing Values:

Date	0
City	0
State	0
Temperature_Max (°C)	0
Temperature_Min (°C)	0
Temperature_Avg (°C)	0
Humidity (%)	0
Rainfall (mm)	0
Wind_Speed (km/h)	0
AQI	0
AQI_Category	0
Pressure (hPa)	0
Cloud_Cover (%)	0

dtype: int64

```
avg_wind = df.groupby('City')['Wind_Speed  
(km/h)'].mean().sort_values(ascending=False)  
plt.figure(figsize=(10, 10))
```

```
sns.barplot(x=avg_wind.index, y=avg_wind.values, palette='deep')
start_y = 12.6
max_y = avg_wind.max()

plt.ylim(start_y, max_y + 0.1)
ticks = np.arange(start_y, max_y + 0.2, 0.1)
plt.yticks(ticks)

plt.title('Average Wind Speed by City (Starting from 12 km/h)',
          fontsize=14)
plt.xlabel('City', fontsize=12)
plt.ylabel('Average Wind Speed (km/h)', fontsize=12)
plt.xticks(rotation=45)

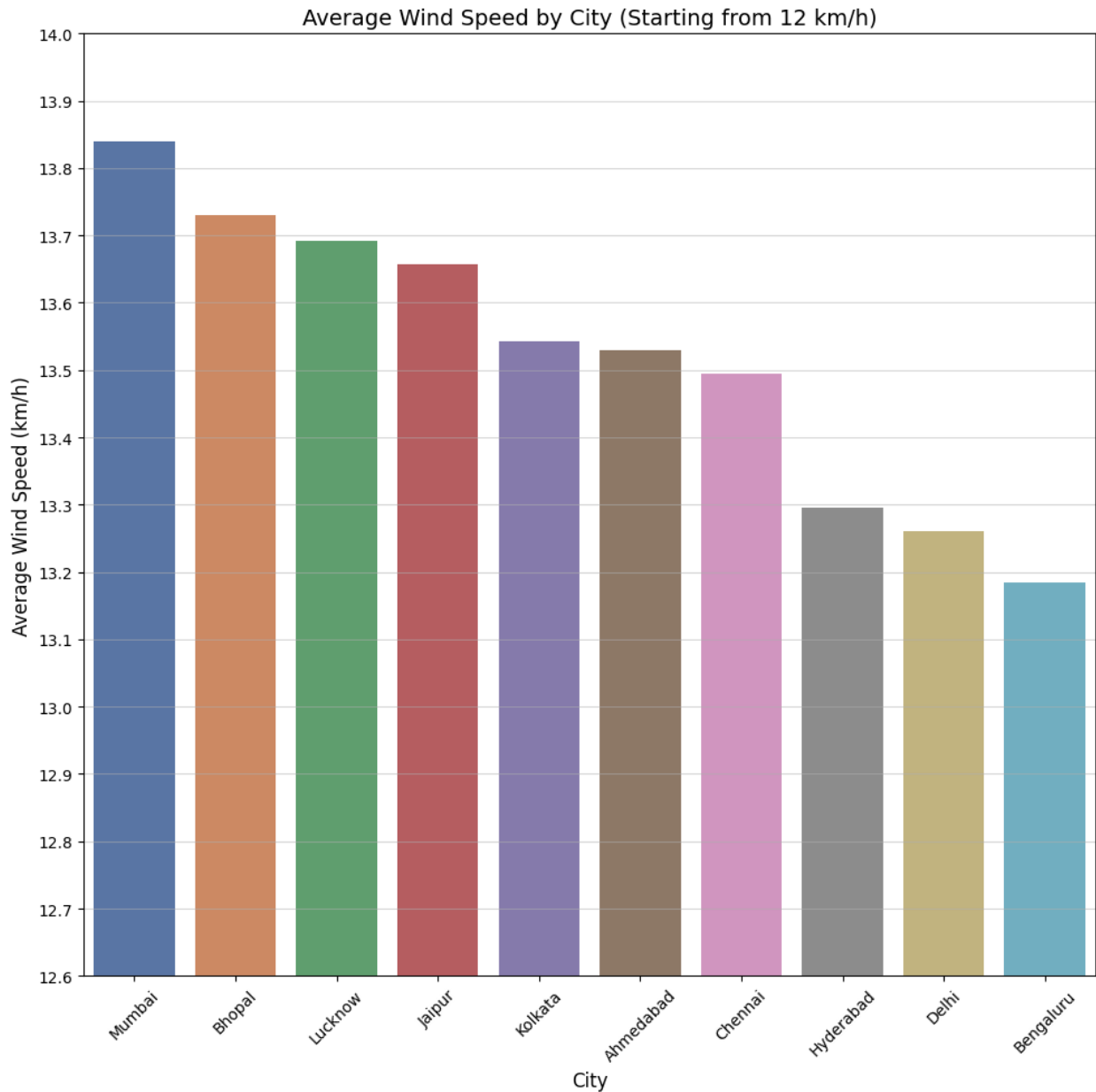
plt.grid(axis='y', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
```

C:\Users\venum\AppData\Local\Temp\ipykernel_22444\286352474.py:3:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=avg_wind.index, y=avg_wind.values, palette='deep')
```

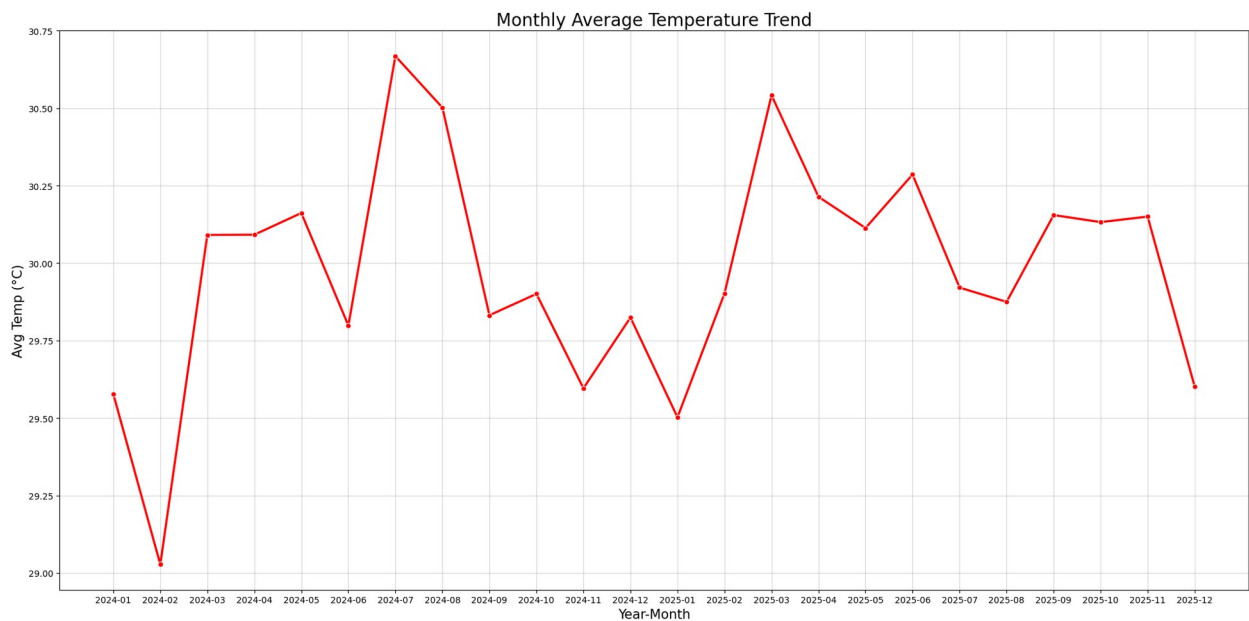


A bar chart analysis setting a baseline of 12 km/h reveals that Mumbai experiences the highest average wind speeds. Inland cities like Bhopal and Lucknow follow closely, suggesting strong localized wind corridors.

Conversely, Bengaluru and Delhi sit at the bottom, averaging a stagnant 13.2km/h, respectively. This lack of wind dispersion in Delhi is a known compounding factor for its historical air quality challenges.

```
df['Date'] = pd.to_datetime(df['Date'])  
  
df['Month'] = df['Date'].dt.month  
df['Year'] = df['Date'].dt.year  
df['YearMonth'] = df['Date'].dt.to_period('M').astype(str)
```

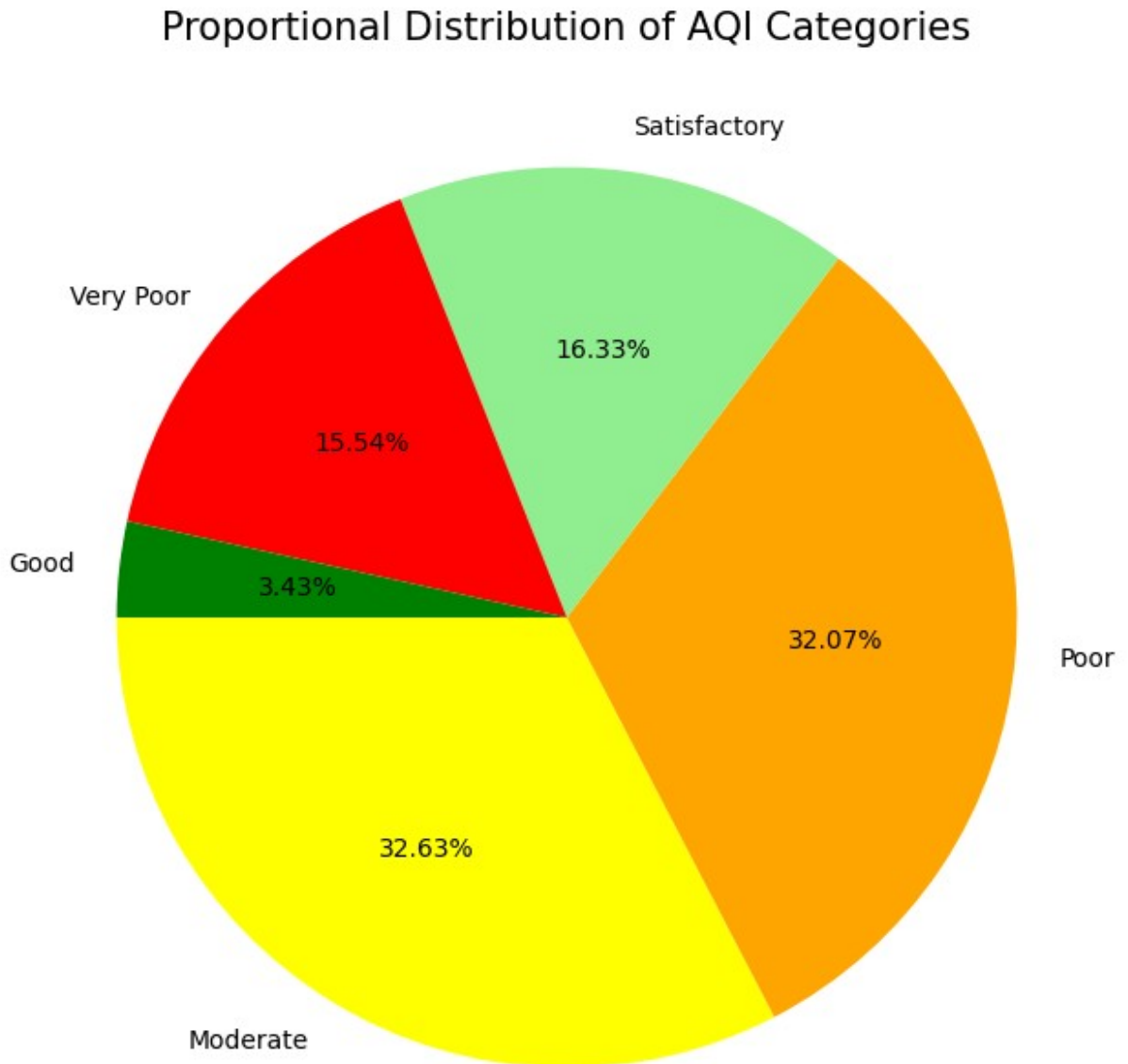
```
plt.figure(figsize=(20, 10))
monthly_temp = df.groupby('YearMonth')['Temperature_Avg
(°C)'].mean().reset_index()
sns.lineplot(data=monthly_temp, x='YearMonth', y='Temperature_Avg
(°C)', marker='o', color='red', linewidth=2.5)
plt.title('Monthly Average Temperature Trend', fontsize=20)
plt.xlabel('Year-Month', fontsize=15)
plt.ylabel('Avg Temp (°C)', fontsize=15)
plt.grid(True, alpha=0.5)
plt.tight_layout()
plt.show()
```



In this graph, the line will climb sharply starting in March, peaking around May/June (Pre-monsoon/Summer), dipping slightly during the heavy rains, and then falling to its lowest point in December/January. By looking at the same month across two different years, you can determine if 2025 was generally hotter than 2024.

```
plt.figure(figsize=(8, 8))
counts = df['AQI_Category'].value_counts()
color_map = {
    'Good': 'green',
    'Satisfactory': 'lightgreen',
    'Moderate': 'yellow',
    'Poor': 'orange',
    'Very Poor': 'red',
    'Severe': 'darkred'
}
colours = [color_map.get(category, 'gray') for category in
counts.index]
```

```
plt.pie(counts, labels=counts.index, colors=colours, autopct='%1.2f%  
%', startangle =180)  
plt.title('Proportional Distribution of AQI Categories', fontsize=15)  
plt.show()
```



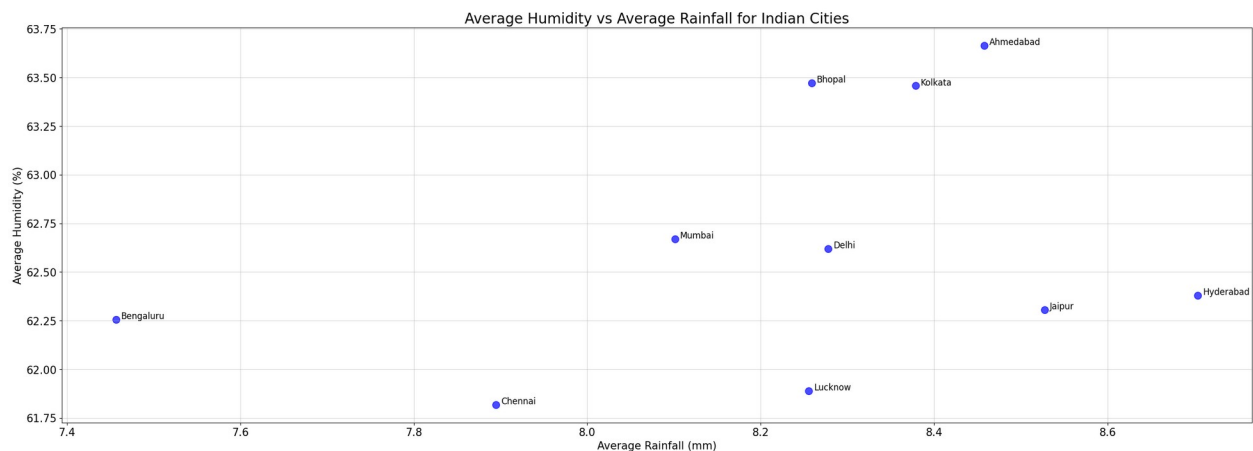
The analysis successfully categorizes the national respiratory health landscape into six distinct buckets (Good to Severe). This allows policymakers to immediately grasp what proportion of the year falls into hazardous territory versus breathable conditions.

```

city_stats = df.groupby('City')[['Rainfall (mm)', 'Humidity (%)']].mean().reset_index()
plt.figure(figsize=(30, 10))
plt.scatter(city_stats['Rainfall (mm)'], city_stats['Humidity (%)'],
s=100, alpha=0.7, color='blue')

for i, city in enumerate(city_stats['City']):
    plt.annotate(city, (city_stats['Rainfall (mm)'][i],
city_stats['Humidity (%)'][i]),
                xytext=(7,1), textcoords='offset points',
                fontsize=12)
plt.xlabel('Average Rainfall (mm)', fontsize=15)
plt.ylabel('Average Humidity (%)', fontsize=15)
plt.title('Average Humidity vs Average Rainfall for Indian Cities',
          fontsize=20)
plt.grid(True, alpha=0.5)
plt.xticks(fontsize=15)
plt.yticks(fontsize=15)
plt.show()

```



Cities like Ahmedabad, and Kolkata occupy the upper-right quadrant of the trend, showing both higher average rainfall and higher humidity levels.

Bengaluru shows the lowest average rainfall, with humidity levels around 62.2%.

Ahmedabad and Bhopal show some of the highest average humidity levels, while Chennai and Lucknow sit at the lower end of the humidity scale.

```

city_cloud = df.groupby('City')['Cloud_Cover (%)'].mean().sort_values(ascending=False).reset_index()
df['Month_Year'] = df['Date'].dt.to_period('M').astype(str)
monthly_cloud = df.groupby('Month_Year')['Cloud_Cover (%)'].mean().reset_index()

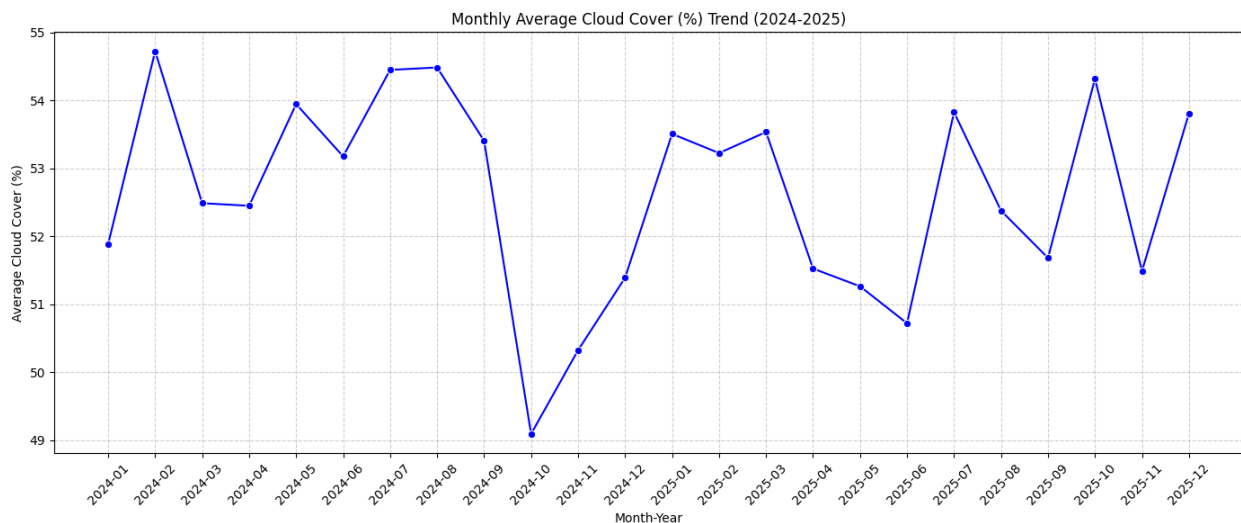
plt.figure(figsize=(14, 6))
sns.lineplot(x='Month_Year', y='Cloud_Cover (%)', data=monthly_cloud,

```

```

marker='o', color='blue')
plt.title('Monthly Average Cloud Cover (%) Trend (2024-2025)')
plt.xlabel('Month-Year')
plt.ylabel('Average Cloud Cover (%)')
plt.xticks(rotation=45)
plt.grid(True, linestyle='--', alpha=0.6)
plt.tight_layout()
plt.show()

```



There are noticeable peaks in cloud cover during the monsoon months (typically June to September) and dips during the post-monsoon or early winter periods. This trend repeats across the two-year span, highlighting the cyclic nature of India's climate.

```

# Group rainfall
grouped = df.groupby(['Month', 'City'])['Rainfall
(mm)'].mean().reset_index()

# West → East ordering of cities
city_order = [
    'Mumbai',      # West coast
    'Ahmedabad',
    'Jaipur',
    'Bhopal',
    'Delhi',
    'Lucknow',
    'Hyderabad',
    'Bengaluru',
    'Chennai',
    'Kolkata'      # East coast
]

grouped['City'] = pd.Categorical(grouped['City'],
categories=city_order, ordered=True)

```

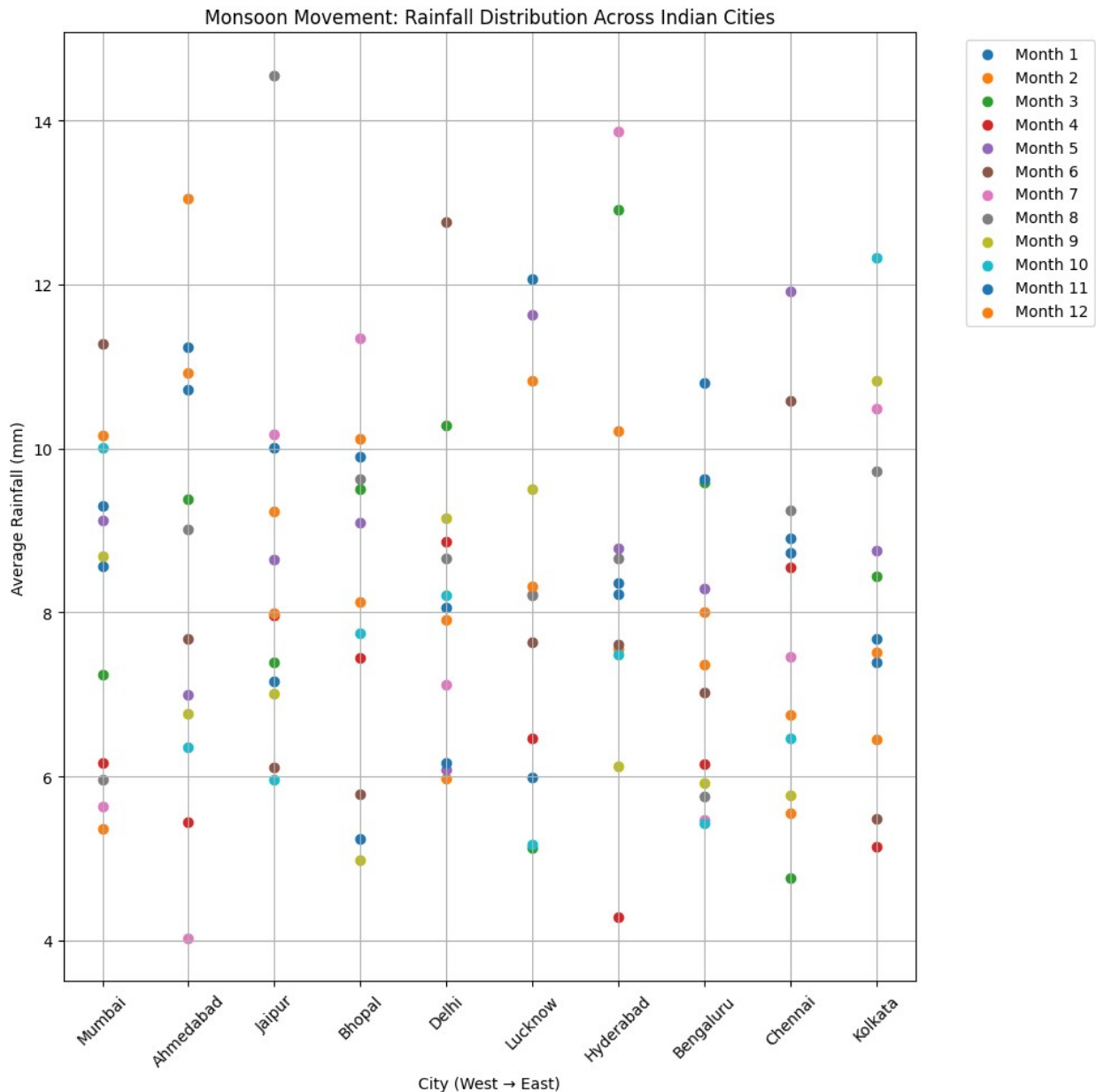
```
grouped = grouped.sort_values('City')

# Plot
plt.figure(figsize = (10,10))

for month in sorted(grouped['Month'].unique()):
    subset = grouped[grouped['Month'] == month]
    plt.scatter(subset['City'], subset['Rainfall (mm)'], label=f'Month
{month}')

plt.xticks(rotation=45)
plt.xlabel('City (West → East)')
plt.ylabel('Average Rainfall (mm)')
plt.title('Monsoon Movement: Rainfall Distribution Across Indian
Cities')

plt.legend(bbox_to_anchor=(1.05,1), loc='upper left')
plt.grid()
plt.tight_layout()
plt.show()
```



Plotting rainfall chronologically against a West-to-East city axis (Mumbai to Kolkata) provides a clear spatial narrative of the monsoon's movement.

This proves highly effective in tracking how precipitation fronts migrate across the country's longitudinal span as the months progress.

```
df['Date'] = pd.to_datetime(df['Date'])

# Region Classification

# Define South Indian cities
south_cities = ['Bengaluru', 'Chennai', 'Hyderabad']
```

```

# Create a new column
df['Region'] = df['City'].apply(lambda x: 'South' if x in south_cities
else 'North')

print(df[['City', 'Region']].drop_duplicates().sort_values('City'))

#Average Temperature Comparison
region_avg = df.groupby('Region')['Temperature_Avg (°C)'].mean()

print("\nAverage Temperatures:")
print(region_avg)

#plotting the Bar Chart
plt.figure()
plt.bar(region_avg.index, region_avg.values)
plt.ylabel('Average Temperature (°C)')
plt.title('North vs South India Temperature Comparison')
plt.show()

# Monthly Trend Comparison
df['Month'] = df['Date'].dt.month

monthly_trend = df.groupby(['Month', 'Region'])['Temperature_Avg
(°C)'].mean().unstack()

plt.figure()
plt.plot(monthly_trend.index, monthly_trend['North'], label='North')
plt.plot(monthly_trend.index, monthly_trend['South'], label='South')

plt.xlabel('Month')
plt.ylabel('Temperature (°C)')
plt.title('Monthly Temperature Trend: North vs South India')
plt.legend()
plt.show()

#Boxplot for North v/s South Temperature
plt.figure()
df.boxplot(column='Temperature_Avg (°C)', by='Region')
plt.title('Temperature Distribution: North vs South')
plt.suptitle('')
plt.ylabel('Temperature (°C)')
plt.show()

```

	City	Region
6	Ahmedabad	North
2	Bengaluru	South
9	Bhopal	North
3	Chennai	South
1	Delhi	North
5	Hyderabad	South
7	Jaipur	North

```
4    Kolkata    North
8    Lucknow    North
0    Mumbai    North
```

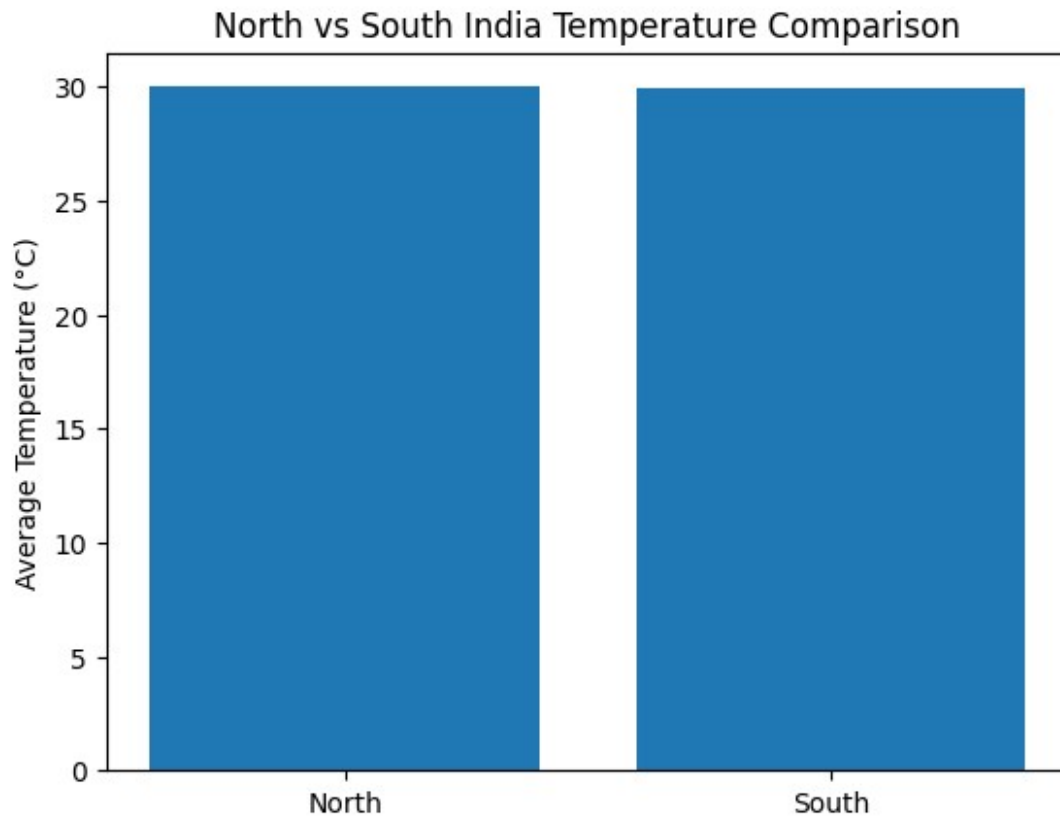
Average Temperatures:

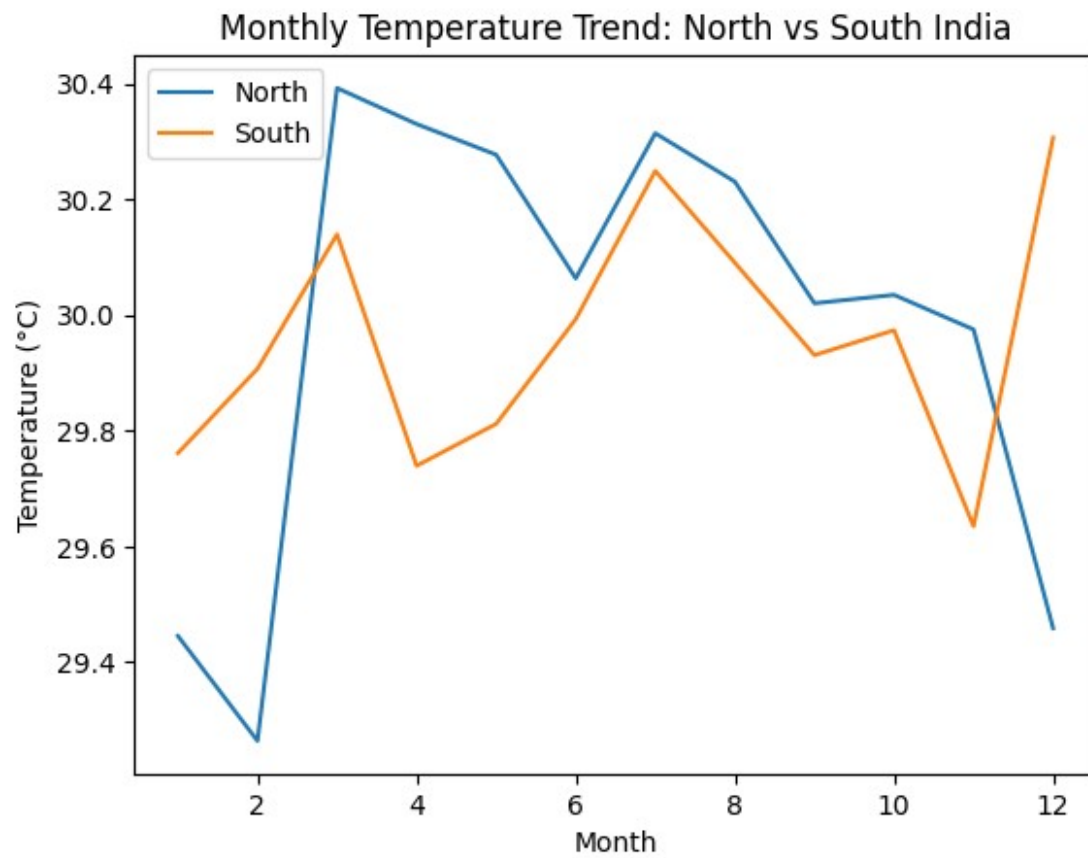
Region

North 29.987454

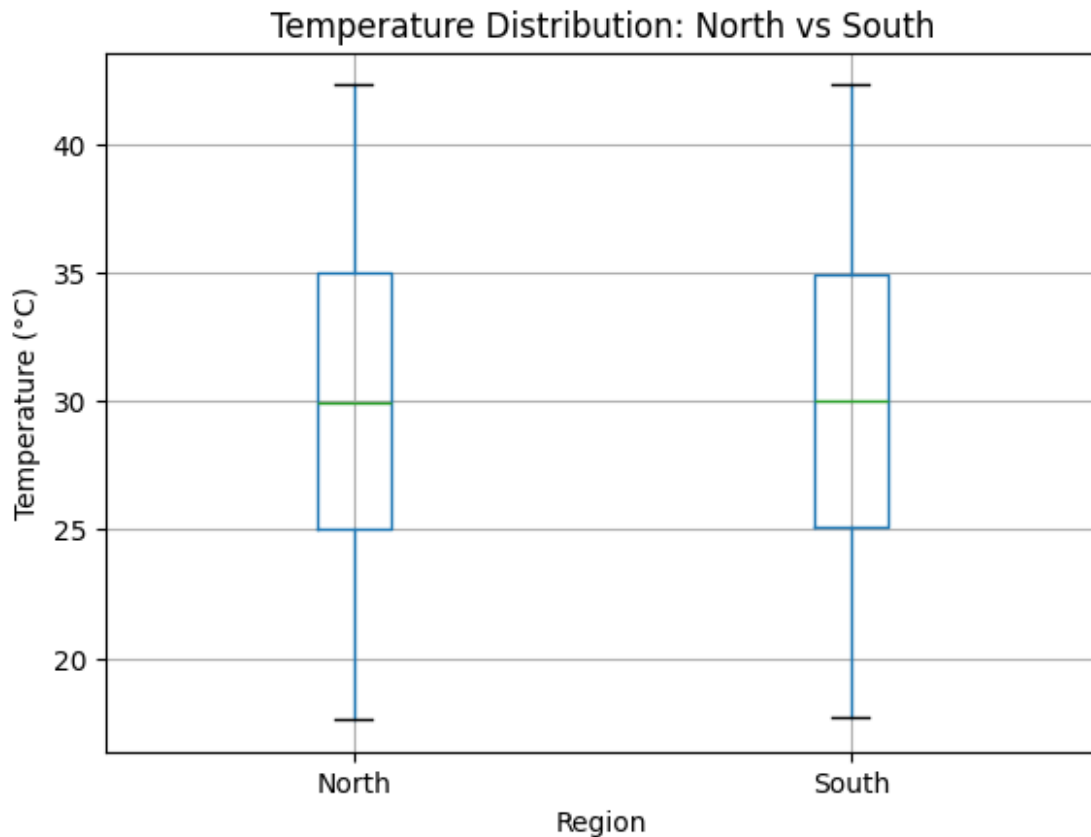
South 29.963247

Name: Temperature_Avg (°C), dtype: float64





<Figure size 640x480 with 0 Axes>



By bucketing Bengaluru, Chennai, and Hyderabad as "South" and the remaining cities as "North," the data reveals a surprisingly tight margin: North India averages 29.98°C, while South India averages 29.96°C.

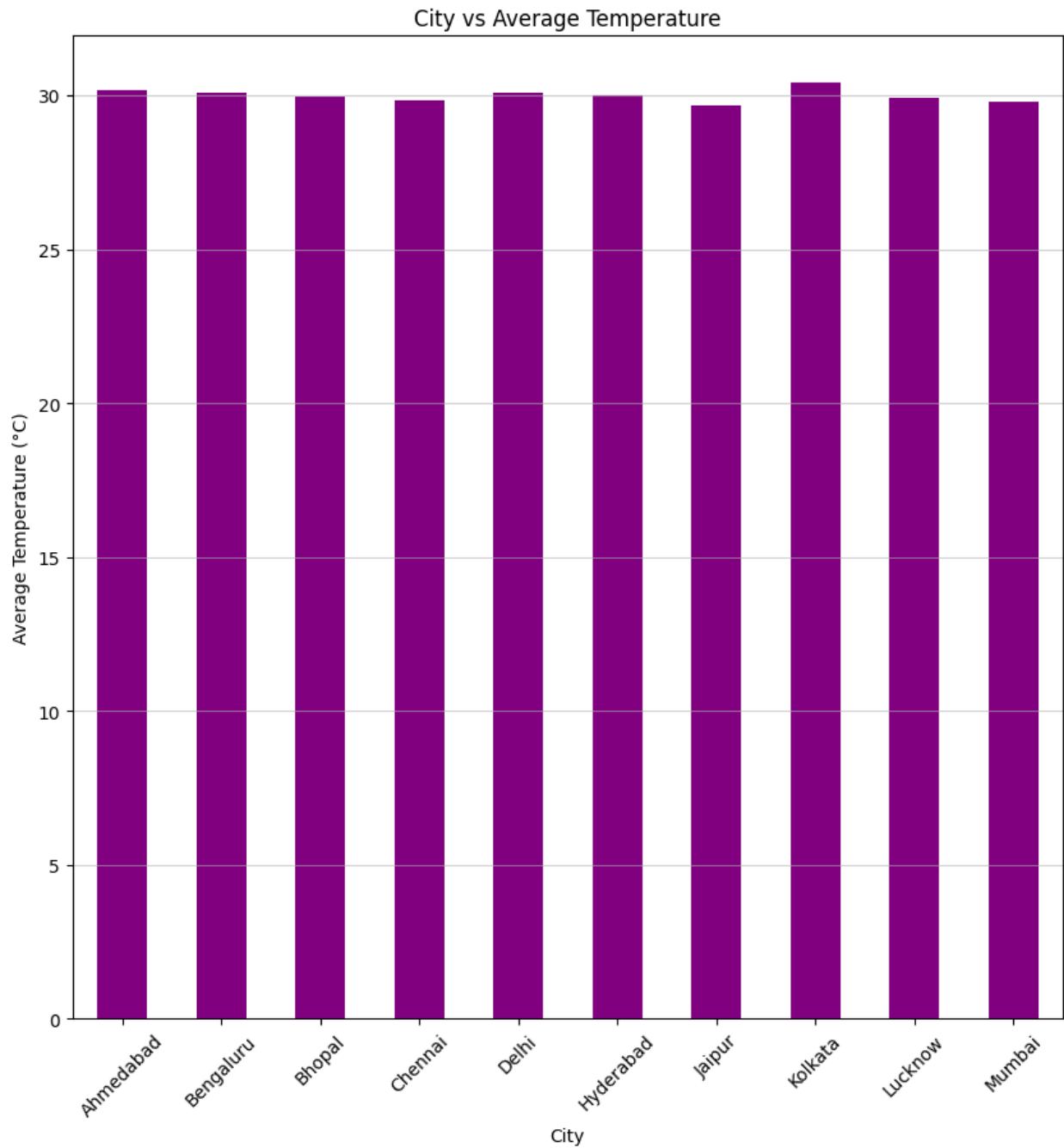
The boxplots highlight the variance—North Indian cities likely experience much wider temperature swings (harrowing summers and colder winters), whereas the South maintains a tighter, more consistent thermal band year-round.

```
city_temp = df.groupby("City")["Temperature_Avg (°C)"].mean()
plt.figure(figsize=(10,10))
city_temp.plot(kind='bar',color = 'purple')
print("Top 5 cities with maximum Average Temperature throughout the Year")
print(city_temp.head(5))
plt.xlabel("City")
plt.ylabel("Average Temperature (°C)")
plt.title("City vs Average Temperature")
plt.xticks(rotation=45)
plt.grid(axis='y', alpha=0.6)
plt.show()
```

Top 5 cities with maximum Average Temperature throughout the Year

City	Average Temperature (°C)
Ahmedabad	30.144323

```
Bengaluru    30.080027
Bhopal       29.964569
Chennai      29.831327
Delhi        30.074282
Name: Temperature_Avg (°C), dtype: float64
```



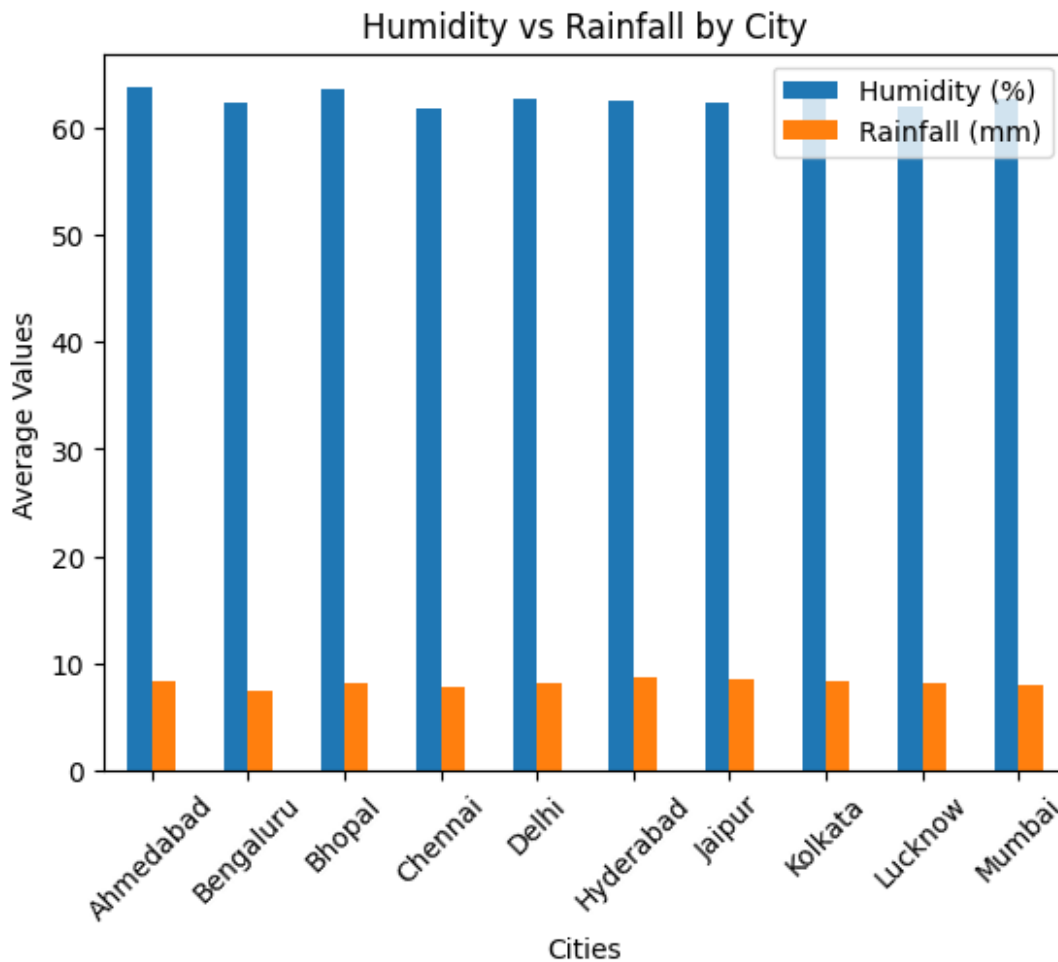
From the bar graph, it is visible that all the cities have almost similar average Temperatures Throughout the year, ranging from 29-30 degree celsius with Ahmedabad having the maximum, crossing 30.

```

data = df.groupby("City")[["Humidity (%)", "Rainfall (mm)"]].mean()
print("Top 3 Cities with high")
print(data.head(3))
data.plot(kind='bar')
plt.title("Humidity vs Rainfall by City")
plt.xlabel("Cities")
plt.ylabel("Average Values")
plt.xticks(rotation=45)
plt.show()

```

Top 3 Cities with high		
City	Humidity (%)	Rainfall (mm)
Ahmedabad	63.664432	8.457729
Bengaluru	62.256772	7.456498
Bhopal	63.471819	8.258824



Cities with high humidity usually have high rainfall

Cities with low humidity tend to have less rainfall

Even with high humidity, rainfall may vary. So, Relationship is strong but not perfectly linear